REDOX REACTION

@ Competishun YT lectures Notes -

- Oxidation Number (or Oxidation State)

It is a hypothetical or real charge on an atom in given species when all the bonds associated with this atom are considered ionic. ON of an atom is always an integer or zero but never fractional. However average O.N of an element may be fraction, zero or -ve.

* Guidelines to determine O.N :-

(1) ON of an element in homonuclear molecule is zero.

eg. ON of O in O_2 = 0

O in O_3 = 0

P in P_4 = 0

S in S_8 = 0

C in C_{60} = 0

(2) Sum of ON of all the elements is equal to total actual charge on the species.

eg. CO_2, ON of C + 2 X ON of O = 0

eg. SO_4^{2-}, ON of S + 4 X ON of O = -2

(3) Following elements always show a constant O.N in compounds

Alkali metals : +1

Alkaline Earth metals : +2

Al, Sc ⟶ +3

Zn, Cd ⟶ +2

F ⟶ -1 (It is most EN element in P.T)

(4) O.N of H in Compounds is usually +1 except in ionic hydrides where it is -1

eg NH_3, H_2SO_4, HNO_3, H_3PO_3, $H_2C_2O_4$, H_2S etc

↳ O.N of H ⟶ +1.

But, NaH, $LiAlH_4$, CaH_2 etc

↳ O.N of H ⟶ -1

eg NaH

$+1 + x = 0$

$x = -1$

(5) O.N of oxygen its compounds is usually -2 except in following cases:

eg. Peroxides (H_2O_2, Na_2O_2, BaO_2, ON of 'O' = -1)

Superoxides (KO_2, O.N of 'O' = $-1/2$)

& Ozonides (KO_3, O.N of 'O' = $-1/3$)

NOTE :

(1) Maximum O.N of an element can be +8.

(2) Range of O.N of p-block elements

Max. ON = Group No − 10

Min ON = Group No − 18

eg. Halogens (G. No = 17) Max. ON = +7 Min. ON = −1	eg. Chalcogens (G. No = 16) Max. ON = +6 Min. O.N = −2
Pnictogens (G. No = 15) Max. ON = +5 Min. ON = −3	Noble Gas (G. No = 18) Max. ON = +8 → eg XeO_4, XeO_3 (+8) (+6) Min. ON = 0

* Range of O.N of d-block elements :

	3	4	5	6	7	8	9	10	11	12
3d Series :	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Max. ON :	+3	+4	+5	+6	+7	+8				

OsO_4 (+8) | eg FeO_4^{2-} (+6)

Max. ON of d-block element = Group No.

Q. Find O.N of underlined elements —

(1) $KMnO_4$, $+1 + x + 4 \times (-2) = 0 \Rightarrow x = +7$

(2) $K_2Cr_2O_7$, $2 \times (+1) + 2x + 7 \times (-2) = 0 \Rightarrow x = +6$

(3) $H_2C_2O_4$, $+2 + 2x - 8 = 0 \Rightarrow x = +3$

(4) Fe_3O_4 , $3x - 8 = 0 \Rightarrow x = +8/3$ (fractional)

$Fe_3O_4 \rightarrow FeO \cdot Fe_2O_3$; FeO: (+2), Fe_2O_3: (+3)

(5) $K_4[Fe(CN)_6]$, $+4 + x + 6 \times (-1) = 0 \Rightarrow x = +2$

(6) H_3PO_4 , $+3 + x - 8 = 0 \Rightarrow x = +5$

(7) H_2SO_4 , $+2 + x - 8 = 0 \Rightarrow x = +6$

(8) H_2SO_5 , $+2 + x - 10 = 0 \Rightarrow x = +8$ (Not possible)

Caro's acid

Correct ON of 'S' in H_2SO_5 = +6

(9) $H_2S_2O_8$, $+2 + 2x - 16 = 0 \Rightarrow x = +7$ (Not possible)

Marshall's acid

Correct → +6

(10) HNO_4 , $+1 + x - 8 = 0 \Rightarrow x = +7$ (Not possible)

Correct → +5

(11) $H_2S_2O_7$, $+2 + 2x - 14 = 0 \Rightarrow x = +6$

(12) $Na_2S_2O_3$, $+2 + 2x - 6 = 0 \Rightarrow x = +2$

(13) $Na_2S_4O_6$, $+2 + 4x - 12 = 0 \Rightarrow x = +2.5$

(Average O.N)

Q. Find O.N

(1) As_2S_3 — (+3), (−2)

(2) Cu_3P — (+1), (−3)

(3) FeS_2 — (+2), (−1)

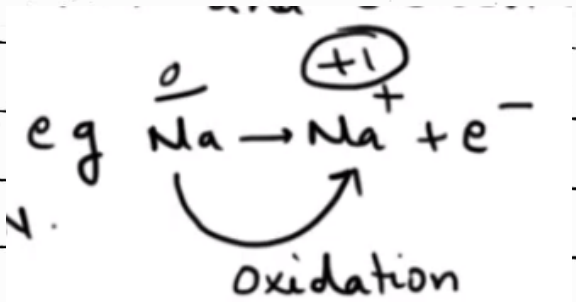
(4) FeC_2O_4 — (+2), (+3), (−2)

(5) $Fe_{0.97}O$

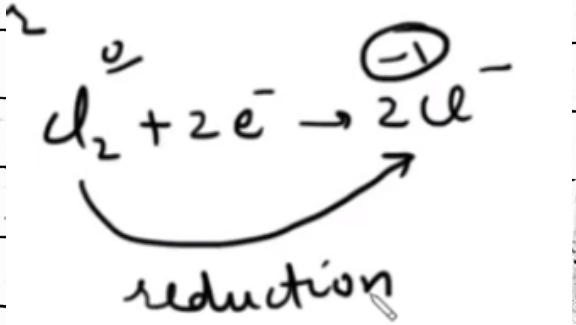
$0.97x - 2 = 0 \Rightarrow x = +\frac{2}{0.97}$ (fractional)

Redox reaction :-

Rex. involving reduction and oxidation.



Oxidation - means (i) losing e^- (or)
(ii) increase in O.N



Reduction - means (i) gaining e^- (or)
(ii) decrease in O.N

Oxidising Agent (oxidant) :

Substance which oxidise others & self get reduced.

eg. $KMnO_4$, $K_2Cr_2O_7$, HNO_3, H_2SO_4, O_2 etc.

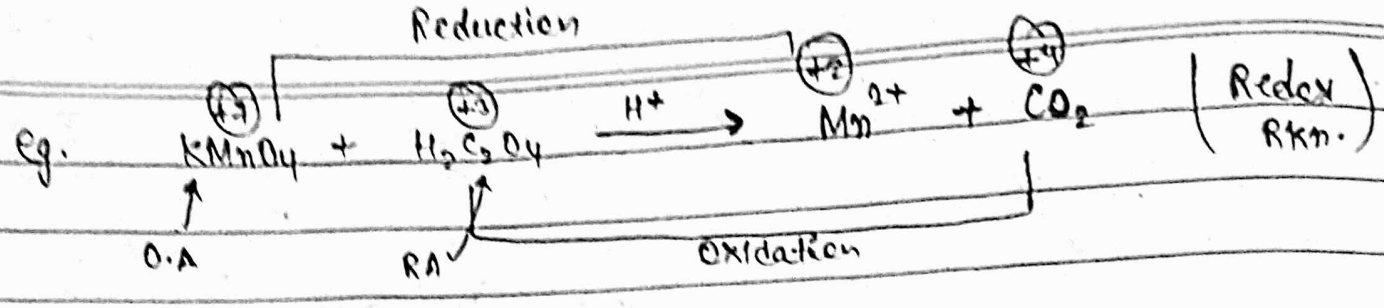
↳ common oxidants.

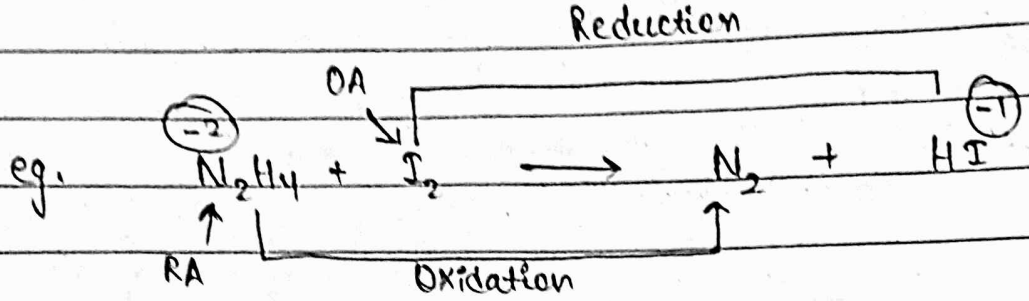
Reducing Agent : (Reductant)

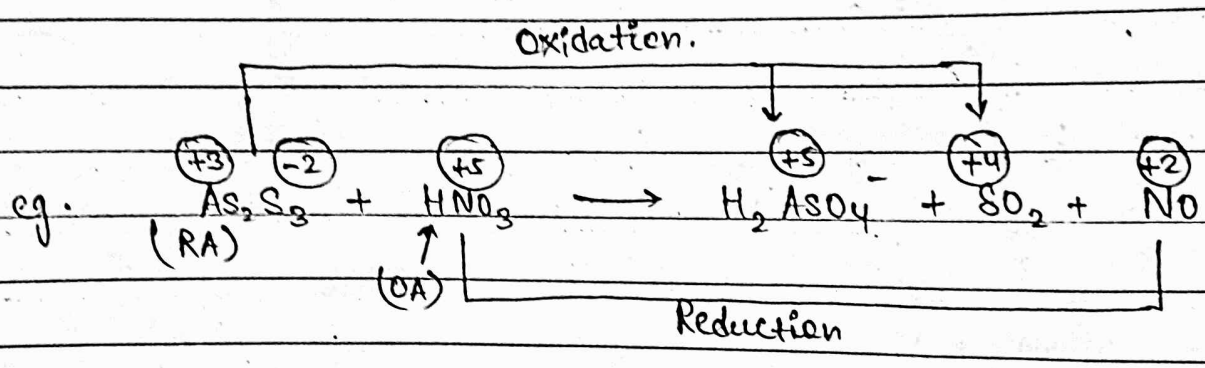
Substance which reduce others & self get oxidised.

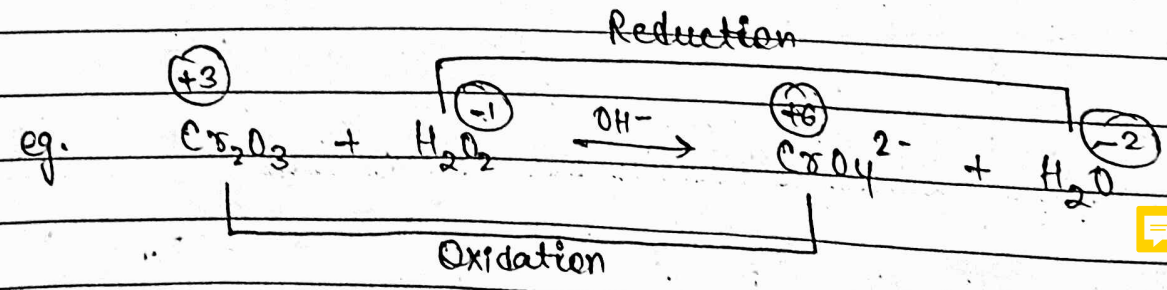
eg. PH_3, HI, $H_2C_2O_4$, CH_4, C, CO, N_2H_4 etc.

Metals (Na, Al, Zn, etc)


Reduction
eg. KMnO4 + H2C2O4 → Mn2+ + CO2
(Redox Rkn.)
O.A
RA
Oxidation

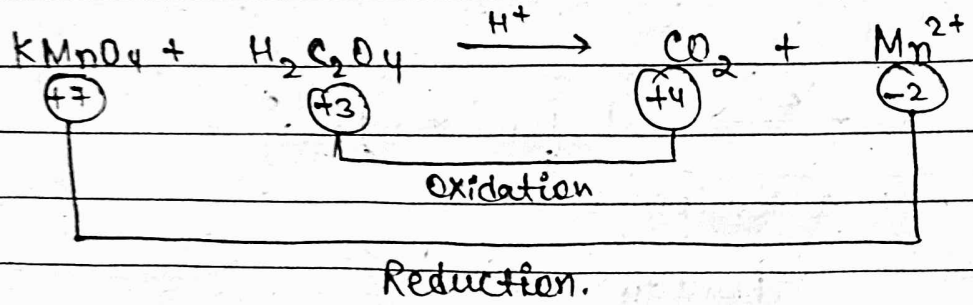

Reduction
OA
eg. N2H4 + I2 → N2 + HI
RA
Oxidation


Oxidation.
eg. As2S3 + HNO3 → H2AsO4- + SO2 + NO
(RA)
(OA)
Reduction

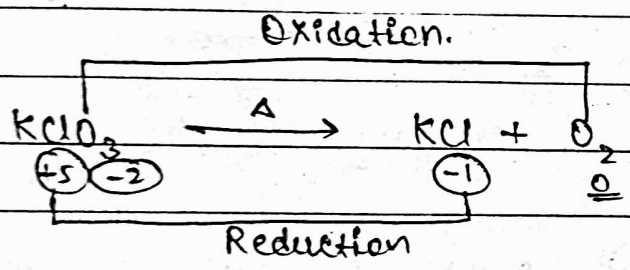

Reduction
eg. Cr2O3 + H2O2 → CrO4 2- + H2O
OH-
Oxidation

* Types of Redox Reactions —

(1) Intermolecular Redox Rxn :-



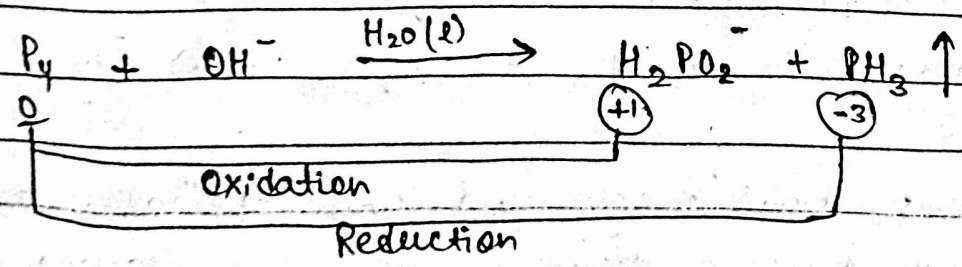
(2) Intramolecular redox reaction :-



(3) Disproportionation Reaction :

Same element from Same Oxidation state from a reactant undergoes oxidation as well as Reduction.

eg.



eg. $Br_2 + OH^- \longrightarrow \underset{(-1)}{Br^-} + \underset{(+5)}{BrO_3^-}$

Reduction

Oxidation.

eg. $\overset{(+3)}{H_3PO_3} \xrightarrow{\Delta} \underset{(+5)}{H_3PO_4} + \underset{(-3)}{PH_3}\uparrow$

Oxidation

Reduction.

* $Br_2 + OH^- \xrightarrow{\text{disproportionation}} Br^- + BrO_3^-$

H^+

Comproportionation

* Comproportionation is the reverse of disproportionation.

(*) Balancing of Redox Reaction :

Ion - Electron Method Oxidation Number Method

(a) Ion - electron Method :

eg. $\underset{(+7)}{KMnO_4} + \underset{(+3)}{H_2C_2O_4} + H_2SO_4 \longrightarrow \underset{(+2)}{MnSO_4} + \underset{(+4)}{CO_2} + K_2SO_4$

Step : (1) Assign Oxidⁿ number and write oxidation Half & Reduction Half in terms of ions and electrons.

$Mn^{(+7)} + 5e^- \longrightarrow Mn^{(+2)}$ (Reduction Half) $\times 2$

$C_2^{(+3)} \longrightarrow 2C^{(+4)} + 2e^-$ (Oxidation Half) $\times 5$

Step (2) :

Add oxidation half and reduction half such that e^- are cancelled out

$$2Mn^{(+7)} + 5C_2^{(+3)} \longrightarrow 2Mn^{(+2)} + 10C^{(+4)}$$

Step (3) : write the balanced Redox part in terms of given species :-

$$2KMnO_4 + 5H_2C_2O_4 \longrightarrow 2MnSO_4 + 10CO_2$$

Step (4) :

Balance elements other than H & O.

$$2KMnO_4 + 5H_2C_2O_4 + 3H_2SO_4 \longrightarrow 2MnSO_4 + 10CO_2 + K_2SO_4$$

Step (5) : (a) In acidic or neutral medium : (H^+ & H_2O should be used)

→ Add H_2O to balance 'O'.

→ Add H^+ to balance 'H'.

$$2KMnO_4 + 5H_2C_2O_4 + 3H_2SO_4 \longrightarrow 2MnSO_4 + 10CO_2 + K_2SO_4 + 8H_2O$$

= Balanced Equation

(b) In Basic Medium : (H_2O and OH^- should be used)

→ Add H_2O to balance 'O'

→ Add H^+ to balance 'H'. Kill H^+ by adding equal number of OH^- to both sides.

Eg. $\underset{(+3)}{Cr(OH)_3} + \underset{(+5)}{IO_3^-} \longrightarrow \underset{(+6)}{CrO_4^{2-}} + \underset{(-1)}{I^-}$ (Basic Medium)

$Cr^{(+3)} \longrightarrow Cr^{(+6)} + 3e^-$ $\times 2$

$I^{(+5)} + 6e^- \longrightarrow I^{\ominus}$ $\times 1$

(+)

$2Cr(OH)_3 + IO_3^- \longrightarrow 2CrO_4^{2-} + I^-$ (Redox part)

$+ 4OH^-$ $+ H_2O + 4H^+ + 4OH^-$

$(2Cr(OH)_3 + IO_3^- + 4OH^- \longrightarrow 2CrO_4^{2-} + I^- + 5H_2O)$

Balanced Equation

Eg. $\underset{(+5)}{NO_3^-} + \underset{0}{Zn} \xrightarrow{OH^-} \underset{(-3)}{NH_3} + \underset{(+2)}{ZnO_2^{2-}}$ (basic medium)

$N^{(+5)} + 8e^- \longrightarrow N^{(-3)}$ $\times (1)$

(+) $Zn \longrightarrow Zn^{2+} + 2e^-$ $\times (4)$

$NO_3^- + 4Zn \longrightarrow NH_3 + 4ZnO_2^{2-}$ (Redox part)

$+ 5H_2O + 7OH^-$ $+ 7H^+ + 7OH^-$

$NO_3^- + 4Zn + 7OH^- \longrightarrow NH_3 + 4ZnO_2^{2-} + 2H_2O$

(b) Oxidation Number Method :-

eg. $NO_3^- + Zn \longrightarrow NH_3 + ZnO_2^{2-}$

(+5) (0) (-3) (+2)

8 2

4 : 1

$NO_3^- + 4Zn \longrightarrow NH_3 + 4ZnO_2^{2-}$

eg.

$Cr(OH)_3 + IO_3^- \longrightarrow CrO_4^{2-} + I^-$ (Basic)

(+3) (+5) (+6) (-1)

3 6

1 : 2

$2Cr(OH)_3 + IO_3^- \longrightarrow 2CrO_4^{2-} + I^-$ (Redox part is Ready)

gm

eg. $K_2Cr_2O_7 + H_2O_2 + HCl \longrightarrow CrCl_3 + O_2 + KCl$

(+6) (-1) (+3) 0

6 2

3 : 1

$K_2Cr_2O_7 + 3H_2O_2 + 8HCl \longrightarrow 2CrCl_3 + 3O_2 + 2KCl + 7H_2O$

eg. $Fe_3O_4 + MnO_4^- \xrightarrow{OH^-} Fe_2O_3 + MnO_2$

(Fe_3O_4: +8/3; MnO_4^-: +7; Fe_2O_3: +3; MnO_2: +4)

1 | 3

$3Fe_3O_4 + MnO_4^- \longrightarrow \frac{9}{2} Fe_2O_3 + MnO_2$

$6Fe_3O_4 + 2MnO_4^- \longrightarrow 9Fe_2O_3 + 2MnO_2 + 2OH^- + H_2O$
$+ 2H^+ + 2OH^-$

$6Fe_3O_4 + 2MnO_4^- + H_2O \longrightarrow 9Fe_2O_3 + 2MnO_2 + 2OH^-$

(Balanced).

eg. $FeC_2O_4 + MnO_4^- \xrightarrow{H^+} Fe^{3+} + CO_2 + Mn^{2+}$

(FeC_2O_4: +2, +3; MnO_4^-: +7; Fe^{3+}: +3; CO_2: +4; Mn^{2+}: +2)

1+2 | 3 | 5

$5FeC_2O_4 + 3MnO_4^- \longrightarrow 5Fe^{3+} + 10CO_2 + 3Mn^{2+} + 12H_2O$
$+ 24H^+$

Q. $Cu_3P + NO_3^- \xrightarrow{H^+} Cu^{2+} + H_2PO_4^- + NO$

(Cu_3P: +1, -3; NO_3^-: +5; Cu^{2+}: +2; $H_2PO_4^-$: +5; NO: +2)

3+8 = 11 | 3

$3Cu_3P + 11NO_3^- \longrightarrow 9Cu^{2+} + 3H_2PO_4^- + 11NO + 10H_2O$
$+ 26H^+$

Q. Select disproportionation Reaction(s)

(A) $NH_4NO_3 \longrightarrow N_2O + H_2O$ $\quad N^{-3}H_4^{+}\ N^{+5}O_3^{-}$

(B) $S + NaOH \longrightarrow Na_2S + Na_2S_2O_3$

(C) $H_2O_2 \longrightarrow H_2O + O_2$

(D) $AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3$

Ans : B, C

(*) Equivalent Concept —

An alternative method to solve stoichiometric problems without balancing the chemical reaction.

$$\text{No. of gram eq.} = \frac{\text{mass of substance}}{\text{Equivalent mass of substance.}}$$

$$\text{Equivalent mass of substance} = \frac{\text{Molar mass of sub.}}{\text{valency factor (v.f)}}$$

$$\text{No. of g. eq.} = \frac{\text{mass of sub.}}{\text{molar mass}} \times \text{v.f}$$

$$\text{No. of g. eq.} = \text{No. of moles} \times \text{v.f}$$

Normality = $\frac{\text{No. of g. eq of solute}}{\text{volume of solution in lit}}$

$= \frac{\text{No. of moles}}{\text{volume of sol}^n} \times \text{V.f}$

Normality = Molarity × V.f

Mole Concept — Equivalent Concept

No. of moles, Molar mass, Molarity → (V.f) → No. of gram eq., Equivalent mass, Normality

Imp * Determination of eq. wt. or V.f

$\text{Eq. Wt (E)} = \frac{\text{Mol. wt (M)}}{\text{V.f}}$

(1) Eq. wt of an element in oxide = mass of element which combines or displaces 8 g of 'O'.

Eq. wt of an element in chloride = mass of element which combines or displaces 35.5 g of 'Cl'

Eq. wt of an element in hydride = mass of the element which combines or displaces 1 g of 'H'

eg. MgO Eq wt. (E) of Mg in MgO = 12 g/eq

12g 16g

eg. $MgCl_2$ Eq. wt (E) of Mg in $MgCl_2$ = 12 g/eq

24g 71g

eg. MgH_2 Eq. wt (E) of Mg in MgH_2 = 12 g/eq.

24g 2g

Note: In oxides, E of 'O' = 8 g/eq
In hydrides, E of 'H' = 1 g/eq
In chlorides E of 'Cl' = 35.5 g/eq.

Q. A metal (M) oxide has 32% by mass of oxygen. Find eq. wt (E) of metal M.

Soln:

Metal oxide

Metal = 68 g O → 32g

∵ 32 g of 'O' → 68 g of metal

8 g of 'O' → $\frac{68}{32} \times 8$ = 17 g/eq

E metal = 17 g/eq Ans.

(2) For an ionic compound (or an ion):
(when ionic compound is not involved in any reaction)

V.f of an ion = No. of unit of charge on ion.

eg. Na^+, V.f = 1	eg. Br^-, V.f = 1
Al^{3+}, V.f = 3	SO_4^{2-}, V.f = 2

V.f of an ionic compound = Total +ve charge or total -ve charge per formula unit

eg. NaCl, V.f = 1
$BaCl_2$, V.f = 2
$Al_2(SO_4)_2$, V.f = 6

Note: Eq. wt of an ionic compound = Eq. wt of cation + Eq. wt of anion

eg. CaO, E_{CaO} = Eq. of Ca + Eq. wt of O
= 20 + 8
= 28

(3) V.f of acids or Bases (involved in acid-base reaction only)

Acid : HCl, HNO_3, H_2SO_4, H_3PO_4, CH_3COOH, exalic acid ($H_2C_2O_4$)

Basicity of Acids → Monobasic acids (or monoprotic acid)
eg. HCl, HBr, HNO_3, $HClO_4$, CH_3COOH etc

→ Dibasic acid
eg. H_2SO_4, $H_2C_2O_4$, H_2CO_3

→ Tribasic acid (H_3PO_4 etc.)

Base : $NaOH$, KOH, $RbOH$, $Ba(OH)_2$, $Ca(OH)_2$ etc.

Acidity of Base → Monoacidic Base :
eg. $NaOH$, KOH, NH_4OH

→ Diacidic Base :
eg. $Ca(OH)_2$, $Ba(OH)_2$ etc.

V.f of an acid in an acid-base reaction
= actual no. of H^+ lost or OH^- gained per molecule of acid

V.f of a base in an acid-base reaction
= actual no. of OH^- lost or H^+ gained per molecule of Base.

eg. $HCl \xrightarrow{NaOH} NaCl$

HCl — $vf = 1$, $E_{HCl} = \frac{1 + 35.5}{1} = 36.5$

eg. $H_2SO_4 \xrightarrow{NaOH} NaHSO_4$, $vf = 1$

$H_2SO_4 \xrightarrow{NaOH} Na_2SO_4$, $vf = 2$

eg. $H_3PO_4 \xrightarrow{NaOH} H_2PO_4^-$, $vf = 1$

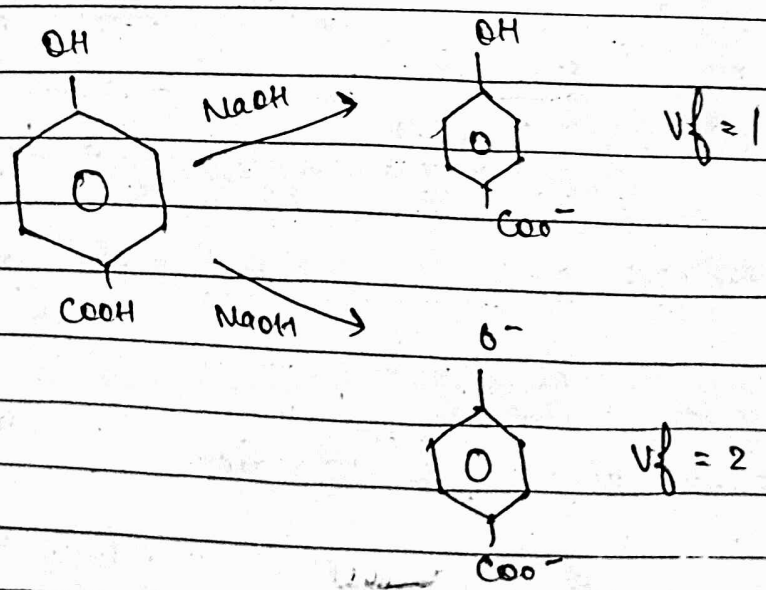
$H_3PO_4 \xrightarrow{NaOH} HPO_4^{2-}$, $vf = 2$

$H_3PO_4 \xrightarrow{NaOH} PO_4^{3-}$, $vf = 3$

eg. $Ba(OH)_2 \xrightarrow{H^+} Ba(OH)^+$, $vf = 1$

$Ba(OH)_2 \xrightarrow{H^+} Ba^{2+}$, $vf = 2$

eg.



$vf = 1$

$vf = 2$

eg. $Al(OH)_3$

- HCl → $Al(OH)_2Cl$, $V_f = 1$
- HCl → $Al(OH)Cl_2$, $V_f = 2$
- HCl → $AlCl_3$, $V_f = 3$
- NaOH → $[Al(OH)_4]^-$, $V_f = 1$

eg. B

Na_2CO_3 (Basic salt)

- HCl → $NaHCO_3$, $V_f = 1$
- HCl → H_2CO_3 ($H_2O + CO_2\uparrow$) , $V_f = 2$

(4) In Redox reaction :

(a) In Half reaction :

→ Change in Oxidation Number per molecule is V.f of an oxidising agent or reducing agent.

eg V.Imp

$KMnO_4$ (+7)

- acidic → Mn^{2+} (+2) , $V_f = 5$, $E = M/5$
- basic or neutral → MnO_2 (+4) , $V_f = 3$, $E = M/3$
- Strongly Basic → MnO_4^{2-} (+6) , $V_f = 1$, $E = M/1$

eg. $\overset{(+6)}{K_2Cr_2O_7} \longrightarrow \overset{(+3)}{Cr^{3+}}$

$K_2Cr_2O_7$: $vf = 6$; Cr^{3+}: $vf = 3$

eg. $\overset{(-2)}{N_2H_4} \longrightarrow \overset{0}{N_2}$

$vf = 4$; $vf = 4$

eg. $\overset{(+5)}{HNO_3} \longrightarrow \overset{(-3)}{NH_3}$

$vf = 8$; $vf = 8$

eg. $\overset{(+2)}{SnCl_2} + \overset{0}{I_2} \longrightarrow \overset{(+4)}{SnCl_4} + \overset{(-1)}{I^-}$

$vf = 2$; $vf = 2$; $vf = 2$; $vf = 2$

eg. $Fe_{0.97}O \longrightarrow Fe^{3+}$

$Fe_{0.97}O$: $\frac{+2}{0.97}$; Fe^{3+}: $(+3)$

$\left(3 - \frac{2}{0.97}\right) \times 0.97 = vf$

$vf = 2.91 - 2$

$= 0.91$

$\left(3 - \frac{2}{0.97}\right) \times 1$

$vf = \frac{0.91}{0.97}$

eg. $Fe_3O_4 \longrightarrow Fe_2O_3$

Fe_3O_4: $(+8/3)$; Fe_2O_3: $(+3)$

$\left(3 - \frac{8}{3}\right) \times 3 = vf$

$vf = 1$

$\left(3 - \frac{8}{3}\right) \times 2 = vf$

$\frac{2}{3} = vf$

eg. $FeC_2O_4 \xrightarrow{OA} Fe^{3+} + CO_2$

(+2) (+3) (+3) (+4)

vf = 1+2 = 3 vf = 1 vf = 1

eg. $As_2S_3 \xrightarrow{OA} H_2AsO_4^- + SO_4^{2-}$

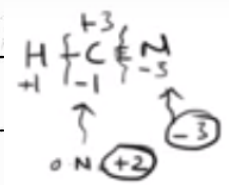
(+3) (-2) (+5) (+6)

vf = 4+24 = 28 vf = 2 vf = 8

eg. $K_4[Fe(CN)_6] \xrightarrow{OA} Fe^{3+} + CO_2 + NO_3^-$

(+2) (+3) (-3) (+3) (+4) (+5)

vf = 61 vf = 1 vf = 2 vf = 8



eg. $Ba(SCN)_2 \longrightarrow Ba^{2+} + SO_4^{2-} + HCN$

(-2) (+4) (-3) (+6) (+2) (-3)

vf = 10+40 = 50 (8-2) x2

= 12 for $Ba(SCN)_2$

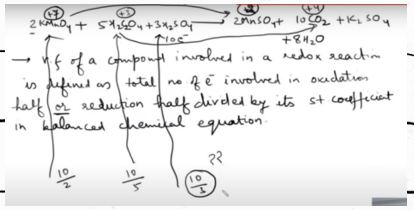
(b) for complete Redox Reaction :-

eg. $\overset{+7}{KMnO_4} + \overset{+3}{H_2C_2O_4} + H_2SO_4 \longrightarrow \overset{+2}{MnSO_4} + \overset{+4}{CO_2} + K_2SO_4$

$Vf = 5 \qquad Vf = 2$

$\overset{+7}{2KMnO_4} + 5H_2C_2O_4 + 3H_2SO_4 \longrightarrow \overset{+2}{2MnSO_4} + 10CO_2 + K_2SO_4 + 8H_2O$ (Balanced)

→ Valency factor (V.f) of a compound involved in a redox reaction is defined as total no. of e^- involved in oxidation half or Reduction half divided by its stoichiometric coefficient in balanced chemical equation.



Determination of Vf of species undergoing disproportionation

$$V.f = \frac{n_1 \times n_2}{n_1 + n_2}, \quad n_1 \rightarrow \text{V.f in oxidation}, \quad n_2 \rightarrow \text{V.f in reduction.}$$

eg. $\underset{0}{Br_2} + Br_2 \xrightarrow{OH^-} \overset{-1}{Br^-} + \overset{+5}{BrO_3^-}$ ($n = 2$, $n_1 = 10$)

$$Vf \text{ of } Br_2 = \frac{2 \times 10}{2+10} = \frac{20}{12} = \frac{5}{3}$$

eg. $\overset{+1}{H_3PO_2} + H_3PO_2 \xrightarrow{\Delta} H_3PO_4 + \overset{-3}{PH_3}$ ($n_1 = 4$, $n_2 = 4$)

$$V.f \text{ of } H_3PO_2 = \frac{4 \times 4}{4+4} = \frac{16}{8} = 2$$

* Law of chemical equivalence –

⊙ $A + B \longrightarrow C + D$ (unbalanced)

No. of gram eq. of each reactants consumed
= No. of g. eq of each product formed.

No. of g. eq. of A reacted = No. of g. eq of B reacted
= No. of g. eq of C formed
= No. of g. eq of D formed.

$$N_1V_1 = N_2V_2$$

$$M_1(V_f)_1 \times V_1 = M_2(V_f)_2 \times V_2$$

$$n_1 \times (V_f)_1 = n_2 \times (V_f)_2$$

Q1 How many moles of $KMnO_4$ in acidic medium are needed to oxidise 100 ml, 0.2 M KNO_2 solution.

Soln $\overset{(+7)}{KMnO_4} + \overset{(+3)}{KNO_2} \xrightarrow{H^+} \overset{(+2)}{Mn^{2+}} + \overset{(+5)}{NO_3^-}$

$V_f = 5 \qquad V_f = 2$

No. of g. eq of $KMnO_4$ = No. of g. eq of KNO_2

$n_{KMnO_4} \times 5 = (0.1 \times 0.2) \times 2$

$\Rightarrow n_{KMnO_4} = \frac{0.04}{5} = \frac{40 \times 10^{-3}}{5} = 8 \times 10^{-3}$ mol.

Q2. 40 ml of 1M Ce^{4+} (ceric ion) is required to oxidise 20ml of 1M $SnCl_2$. find final O.N of Ce.

Soln:

$$\underset{40ml,1M}{Ce^{4+}} + \underset{20ml\ 1M}{Sn^{2+}} \longrightarrow Sn^{4+} + Ce^{x}$$

$(4-x)$; Sn^{2+} → Sn^{4+} : (2)

No. of g. eq of Ce^{4+} = No. of g. eq of Sn^{2+}

$$\frac{40}{1000} \times 1 \times (4-x) = \frac{20}{1000} \times 1 \times 2$$

$$4 - x = 1$$

$$x = 3$$

$\Rightarrow Ce^{3+}$ Ans

Q3 What volume of 0.01M $K_2Cr_2O_7$ in acidic medium is required to oxidise 0.4 mol of ferrous oxalate.

Soln:

$$\underset{(+6)}{K_2Cr_2O_7} + \underset{(+2)(+3)}{FeC_2O_4} \xrightarrow{H^+} Cr^{3+} + Fe^{3+} + CO_2$$

$V_f = 6$; $V_f = 1+2 = 3$

No. of g. eq. of $K_2Cr_2O_7$ = No. of g. eq. of FeC_2O_4

$$V \times 0.01 \times 6 = 0.4 \times 3$$

$$V = \frac{0.40}{0.02} = 20\ lit$$ Ans.

Q. What mass of hydrazine (N_2H_4) will get oxidised by 100 ml, 0.2 M I_2, N_2H_4 converts to N_2.

Soln: $\overset{-2}{N_2H_4} + I_2 \longrightarrow \overset{0}{N_2} + HI$

$vf = 4$ $vf = 2$

No. of g. eq of N_2H_4 = No. of g. eq of I_2

$$\frac{W}{32/8} \times 4 = 0.1 \times 0.2 \times 2$$

$$W = 0.8 \times 0.4$$

$$= 0.32\ g \quad \text{Ans.}$$

Q. A mixture of FeO and Fe_2O_3 is treated with acidified KMnO4 (100ml, 0.3M is used). The resulting solution is treated with Zn dust to convert Fe^{3+} to Fe^{2+}. Now solution is reacted with $K_2Cr_2O_7$ (0.13M, 1 lit) and Fe^{2+} is converted to Fe^{3+}. Find no. of moles of FeO and Fe_2O_3 in original mixture.

Soln: $\overset{+2}{FeO} + \overset{+3}{Fe_2O_3} \xrightarrow[0.1L,\,0.23M]{KMnO_4 + H^+} \underset{(x+2y)}{Fe^{3+}} + \underset{(RA)}{Zn} \longrightarrow Fe^{2+}\ (x+2y)$

x mol, y mol

$Fe^{2+}\ (x+2y) \xrightarrow{K_2Cr_2O_7\ (1L,\,0.13M)} Fe^{3+}$

$x \times 1 = 0.1 \times 0.23 \times 5$

$x = 0.115$ mol (FeO)

$x + 2y \times 1 = 1 \times 0.13 \times 6$

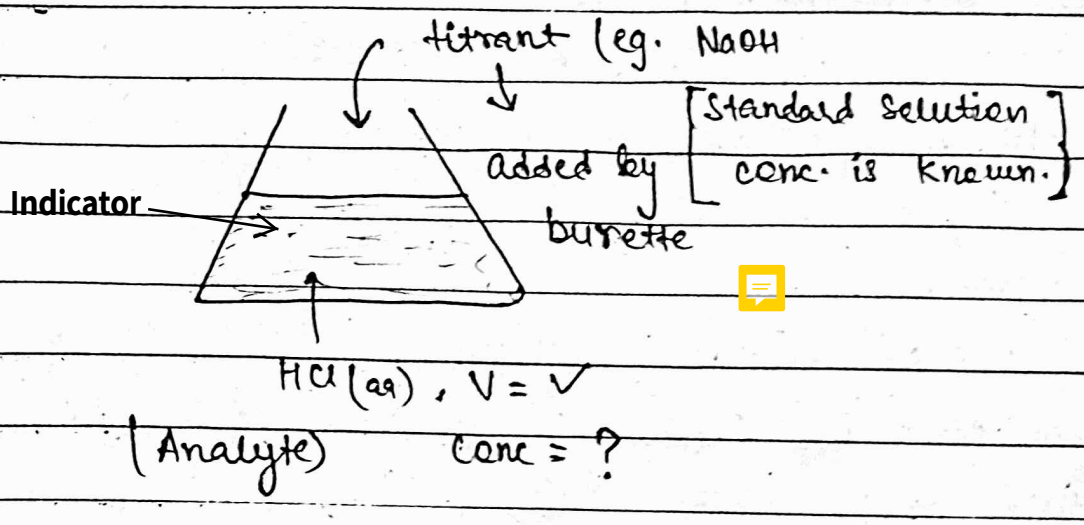
$0.115 + 2y = 0.78$

$y = \frac{0.78 - 0.115}{2}$ mol

(Fe_2O_3)

* Titration :

An experimental method to determine conc. of a solution using a standard solution with the help of a suitable indicator.



$$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O$$

Equivalence point :

A theoretical point when reaction is complete

End point :

Experimental point during titration when indicator shows sharp colour change.

→ For an accurate titration, end point should be as close as possible to the equivalence point.

Q What mass of I_2 is needed to oxidise 200 ml, 0.4N hyposolution ($Na_2S_2O_3$) to convert it into sodium tetra thionate ($Na_2S_4O_6$)

Soln $\overset{(+2)}{I_2} + Na_2S_2O_3 \longrightarrow \overset{(+2.5)}{Na_2S_4O_6} + I^-$

$vf = 2$ $\quad vf = 1$

$$\left(\frac{W}{2\times127}\right)\times 2 = (0.2\times0.4)$$

$$W = 0.04\times2\times127\ g$$

2nd Q A sample consisting of 1 mol each of Fe_3O_4 and $Fe_{0.97}O$ is treated with 0.5 M $KMnO_4$, x lit was needed to oxidise iron oxides. find x.

Soln : $\overset{(+8/3)}{Fe_3O_4} + \overset{(+2/0.97)}{Fe_{0.97}O} \xrightarrow[+H^+]{KMnO_4} \overset{(+3)}{Fe}$

1 mol $\quad$ 1 mol

$vf = 1$ $\quad vf = \left(3-\frac{2}{0.97}\right)\times0.97$

$$vf \approx 0.91$$

$$(1\times1)+(1\times0.91) = 0.5\times x\times5$$

$$x = \frac{1.91}{2.5}\ \text{lit}$$ Ans

* Types of Titrants —

⊙ Primary titrant (or primary standard)
→ whose weight can be accurately weighed & such solution need not be standardised before use.

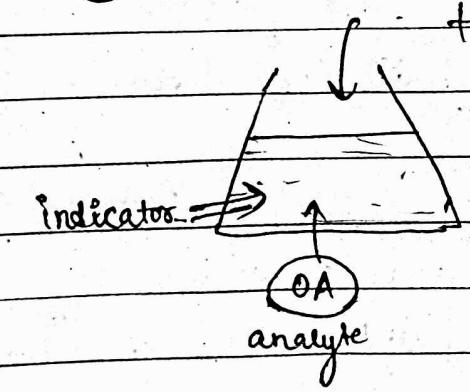
eg. $K_2Cr_2O_7$, Mohr Salt, KCl, $H_2C_2O_4$ etc.

⊙ Secondary titrant (or Secondary standard)
→ whose weight can't be accurately weighed (due to impurities / moisture etc)

→ Need be standardised before use.

eg. $KMnO_4$, $Na_2Cr_2O_7$, NaOH etc.

(1) Redox Titration —



At equivalence point,
No. of g. eq of OA = No. of g. eq of RA

$N_1V_1 = N_2V_2$

Some common oxidising agents and reducing agents involved in Titrations

Titration involving $KMnO_4$: gmf

OA → $KMnO_4$ → acidic → Mn^{2+} (+2)
→ basic or neutral → MnO_2 (+4)
→ Strongly Basic → MnO_4^{2-} (+6)

Remember!

→ Self indicator (due to its intense colour, purple)

imp → It is titrated with Mohr Salt ($FeSO_4(NH_4)_2SO_4 \cdot 6H_2O$)

$$\overset{(+2)}{FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O} + KMnO_4 \xrightarrow{H^+} Mn^{2+} + Fe^{3+} + \boxed{NH_4^+}$$

$vf = 1$ $\qquad$ $vf = 5$

→ To maintain acidic medium of $KMnO_4$, H_2SO_4 is preferred (not to use HCl or HNO_3)

titrant

$$Fe^{2+}(aq) + KMnO_4 \xrightarrow{H^+} Fe^{3+} + Mn^{2+}$$

At eq. point, No. of Eq. of Fe^{2+} = No of eq. of $KMnO_4$

$$N_1V_1 = N_2V_2$$

eg.

Fe^{2+} + $KMnO_4$ — HCl (V_1); HNO_3 (V_2); H_2SO_4 (V_3)

$\boxed{V_1 > V_3 > V_2}$

(2) $K_2Cr_2O_7$:

An oxidising agent used in acidic medium

$$\overset{(+6)}{Cr_2O_7^{2-}}(aq) \underset{H^+}{\overset{OH^-}{\rightleftharpoons}} \overset{(+6)}{CrO_4^{2-}}$$

Orange Red $\qquad$ yellow Soln

$v_f = 6$

→ $Cr_2O_7^{2-}$ $\xrightarrow{RA}$ Cr^{3+}

(OA) (green)

→ $S^{2-} \longrightarrow S$

$SnCl_2 \longrightarrow SnCl_4$

$Fe^{2+} \rightarrow Fe^{3+}$

$C_2O_4^{2-} \rightarrow CO_2$

$AsO_3^{3-} \longrightarrow AsO_4^{3-}$

} by $K_2Cr_2O_7$ (or) $KMnO_4$

* Titration Involving Iodine (I_2) :

(a) Iodimetric titration :

I_2 is used as oxidising Agent and used to titrate reducing agent.

eg. $Sn^{2+} + I_2 \longrightarrow Sn^{4+} + I^-$

eg. $S_2O_3^{2-} + I_2 \longrightarrow S_4O_6^{2-} + I^-$

(hypo : $Na_2S_2O_3$)

eg. $AsO_3^{3-} + I_2 \longrightarrow AsO_4^{3-} + I^-$

eg. $N_2H_4 + I_2 \longrightarrow N_2 + HI$

→ Starch is used as indicator which gives blue colour (or blue/black) with I_2.

→ eg. $I_2 + KI(aq) \longrightarrow KI_3(aq)$ or $I_3(aq)$

I_2 ↑ dark violent colour.

Brown Solution

(b) Iodometric titration :-

Oxidising agents are reacted with excess of solid KI and liberated I_2 or I_3^- is estimated with the help of standard hypo solution ($Na_2S_2O_3$)

eg. $Cu^{2+}(aq) + KI$ (excess) $\xrightarrow{Rex-I}$ $CuI\downarrow + I_2$ or I_3^-

(I_2 : intermediate)

↑ OA

(To determine conc. of Cu^{2+})

I_2 or I_3^- $\xrightarrow{Rex-II}$ $Na_2S_2O_3$ (Standard hypo solution)

$NaI + Na_2S_4O_6$

(Sodium tetra thionate)

No. of g. eq of hypo

= No. of gram eq of I_2 reacted in rex - II

= No. of n_{I_2} in rex - II × 2

n_{I_2} formed in eq I = No of moles of I_2 reacted in eq II.

No. of g. eq. of Cu^{2+} = No. of g. eq of I_2 formed in rex - I

⇒ No. of g. eq of Cu^{2+} = No. of g eq. of hypo

$$N_1V_1 = N_2V_2$$

Q. $KMnO_4$ ($v_f = 3$) + KI $\xrightarrow[\text{Rex-I}]{OH^-}$ MnO_2 + IO_3^-

$KMnO_4$: 100 ml, x molar

IO_3^- $\xrightarrow{\text{Rex-II}}$ SO_2 (g) (0.5 mol of SO_2 is used up)

IO_3^- $\rightarrow$ $I_2 + SO_4^{2-}$

Find x.

Soln: No. of g. eq of SO_2 = No. of g. eq of IO_3^-

For Rkn-II

$0.5 \times 2 = n_{IO_3^-} \times 5$

$n_{IO_3^-} = 0.2$ mol

For Rkn-I: $0.1 \times x \times 5 = 0.2 \times 2$

$x = 4$ molar Ans

Q. IO_3^- + KI (excess) $\xrightarrow{\text{Rkn-I}}$ I_2 ($v_f = 5/2$ → $v_f = 2$)

1 lit, x Molar

I_2 $\xrightarrow{\text{Rkn-II}}$ $Na_2S_2O_3$ (2 lit, 0.2 Molar)

Find x.

$NaI + Na_2S_4O_6$

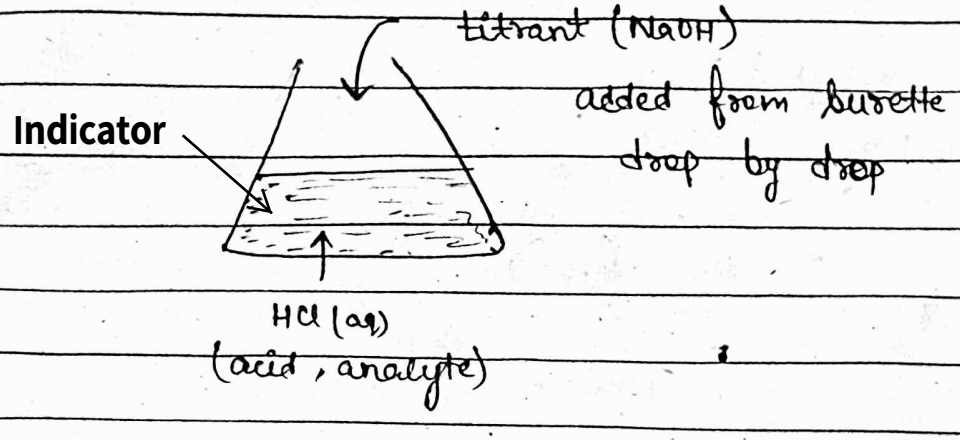
Soln Rkn-II: $n_{I_2} \times 2 = 2 \times 0.2 \times 1$

$n_{I_2} = 0.2$ mol

Rkn-I: $1 \times x \times 5 = 0.2 \times 5/3$

$x = \frac{0.2}{3}$ molar Ans

* Acid- Base titration :-



$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$

At eq. point, No. of g. eq of HCl = No. of g. eq. of NaOH

$$N_1V_1 = N_2V_2$$

* Double Titration :

Purpose is to determine composition of a mixture

eg. $Na_2CO_3 + NaHCO_3$

eg. $NaOH + Na_2CO_3 + NaHCO_3$

→ Two indicators are needed to mark two equivalence point.

→ Most commonly used indicators are HPh (Phenolphthalein) and methyl orange (MeOH)

* HPh (weak acid indicator)

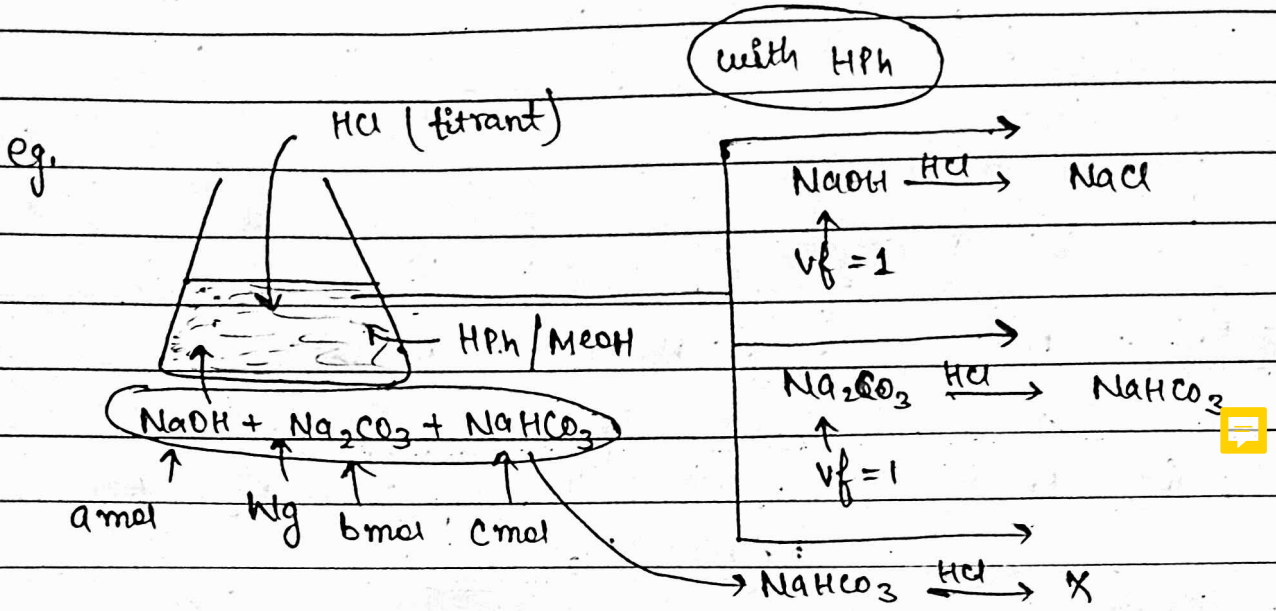
Colourless ← 8 - 10 (pH transition range)

pH → Pink

* MeOH (weak Base indicator)

Yellow ← 4.4 – 3.1 → Red

pH transition Range



$aX40 + bX106 + cX84 = W$ — (1)

At 1st eq. point

$aX1 + bX1 = (\text{No. of g. eq of HCl})_{HPh}$ — (2)

with MeOH (Repeated titration)

→ $NaOH \xrightarrow{HCl} NaCl$, $vf = 1$

→ $Na_2CO_3 \xrightarrow{HCl} H_2CO_3$, $vf = 2$

→ $NaHCO_3 \xrightarrow{HCl} H_2O + CO_2$, $vf = 1$

$$a\times1 + b\times2 + c\times1 = (\text{No. of g. eq of HCl})_{\text{MeOH}} \quad \text{— (3)}$$

Q. A solid sample of Na_2CO_3 and $NaHCO_3$ is dissolved in water to form solution. Half of this solution is titrated with 1 N HCl solution in presence of HPh indicator, 4 ml of HCl (1N) is used up. Then titration is repeated with the other half of solution with same 1N HCl solution but with MeOH indicator this time 11 ml of HCl is used up. Find no. of moles of Na_2CO_3 and $NaHCO_3$ in initial sample.

Sol^n

$$\underset{\uparrow\, x\text{ mol}}{Na_2CO_3} + \underset{\uparrow\, y\text{ mol}}{NaHCO_3}$$

1st equivalence point (with HPh)

$$Na_2CO_3 \longrightarrow NaHCO_3$$

$$\text{No. of g. eq of } Na_2CO_3 = (\text{No. of g. eq of HCl})_{\text{HPh}}$$

$$\frac{x}{2}\times1 = \frac{4}{1000}\times1 \Rightarrow x = 8\times10^{-3}\text{ mol}$$

2nd eq. point (with MeOH),

$$Na_2CO_3 \longrightarrow H_2CO_3\,,\quad NaHCO_3 \longrightarrow H_2CO_3$$

$$\frac{x}{2}\times2 + \frac{y}{2}\times1 = \frac{11}{1000}\times1$$

$$\frac{y}{2} = (11-8)\times10^{-3}\text{ mol}\,,\quad y = 6\times10^{-3}\text{ mol}$$

* Back titration :-

A (analyte) + B (titrant - I, excess) → C

↑ this product C does not react with D.

C + D (another titrant) → E

No. of g. eq of B consumed
= No. of g. eq of A + No. of g. eq of D.

Q. A sample of 100 g of $Ca(OH)_2$ (impure, and impurities do not react) is dissolved 1L, 3M HCl (aq). The remaining HCl is titrated with 1L, 2M NaOH. find % purity of $Ca(OH)_2$, in the sample.

Soln :

$$\underset{W\,g}{Ca(OH)_2} + \underset{\text{(excess)}}{\underset{3\,mol}{HCl}} \longrightarrow CaCl_2 + H_2O$$

$$\underset{2\,mol}{NaOH}$$

$$3 \times 1 = \frac{W}{(40+34)} \times 2 + 2 \times 1$$

$$W = \frac{1 \times 74}{2} = 37 \text{ gram}$$

% $Ca(OH)_2$ in Sample $= \frac{37}{100} \times 100 = 37\%$ Ans.

(*) % of available chlorine in Bleaching powder :

$CaOCl_2$ + Impurities. W_g

Cl-

OCl^- ↑ oxidising anion

+ dil. CH_3COOH / dil HCl / $CO_2(g) + H_2O(l)$

to make medium acidic

$Cl_2 \xrightarrow[\text{excess}]{KI} I_2 + Cl^-$

↑ iodometric titration.

$I_2 \xrightarrow{Na_2S_2O_3} NaS + Na_2S_4O_6$

No of g. eq of Cl_2 = No. of g. eq of hypo

$$\left(\frac{W_{Cl_2}}{71}\right) \times 2 = \left(\frac{V_{Na_2S_2O_3}}{1000}\right) \times N_{Na_2S_2O_3}$$

$$\% \text{ of available chlorine} = \frac{W_{Cl_2}}{W_{sample}} \times 100$$

Q. 3.55 g of bleaching powder sample, is treated with dil HCl and excess of KI. liberated I_2 is estimated with 80 ml, 0.2 molar hypo solution. find % of available chlorine in sample.

Sol^n :

$$\frac{W_{Cl_2}}{71} \times 2 = \frac{80}{1000} \times 0.2 \times 1$$

$$W_{Cl_2} = \frac{0.8 \times 71}{100} \text{ g}$$

% of available Cl_2

$$= \frac{\frac{0.8 \times 71}{100}}{3.55} \times 100 = 16\% \quad \text{Ans.}$$

(*) Hard Water :

Water which does not produce ~~enog~~ enough lather with soaps and detergents, such water is not suitable for laundary purposes. Hardness in water is due to dissolved impurities of salts of Ca^{2+}, Mg^{2+}, Al^{3+} in water.

$$2\,C_{17}H_{35}COONa + Ca^{2+}/Mg^{2+} \longrightarrow (C_{17}H_{35}COO)_2Ca\downarrow + 2Na^{+}$$

(Soap)

* Types of Hardness -

(1) Temporary Hardness -
It is removed by just boiling.

(2) Permanent Hardness -
Can't be removed by just boiling.
Special methods should be employed to remove permanent hardness.

(a) Temporary hardness :
due to bicarbonates of Ca^{2+} and Mg^{2+}

eg. $Ca(HCO_3)_2\,(aq)$, $Mg(HCO_3)_2\,(aq)$

(b) Permanent hardness :
due to dissolved chlorides, sulphates of Ca^{2+}, Mg^{2+}, Al^{3+} etc.

* Methods of removal of hardness of water :

(1) By Boiling :
(for removal of temporary hardness)

$$Mg(HCO_3)_2\,(aq) \xrightarrow{\text{Boil}} \underset{\text{white}}{Mg(OH)_2} + 2CO_2\uparrow$$

$$Ca(HCO_3)_2\,(aq) \xrightarrow{\text{Boil}} \underset{\text{white ppt.}}{CaCO_3\downarrow} + H_2O\,(l) + 2CO_2\uparrow$$

(Solubility of $Mg(OH)_2$ < Solubility of $MgCO_3$)

(2) By using Slaked lime / quick lime : (Clark's Method)
$Ca(OH)_2\,(s)$ / $CaO\,(s)$

(For temporary hardness)

$$Ca(HCO_3)_2\,(aq) + Ca(OH)_2 / CaO\,(s) \longrightarrow 2CaCO_3\downarrow + H_2O$$

(3) By using Washing Soda : (Na_2CO_3)

(for Permanent hardness)

$$CaSO_4(aq) + Na_2CO_3(s) \longrightarrow \underset{\text{ppt.}}{CaCO_3\downarrow} + Na_2SO_4$$

(4) By using ion-exchange resins :-

(for removal of permanent hardness)

(Permutit method)

Resins : Sodium Aluminosilicates

eg. $Na_2Al_2Si_2O_8$ (or) Na_2R

$$Na_2R + Ca^{2+}/Mg^{2+} \longrightarrow \underset{\text{ppt.}}{CaR\downarrow}/\text{or}\ \underset{\text{ppt.}}{MgR\downarrow} + 2Na^+$$

(5) By using polyphosphates : (Calgon method)

→ for permanent hardness.

$(NaPO_3)_n$ → Sodium Polyphosphates

(Soluble in water)

→ In this method Ca^{2+} or Mg^{2+} are sequestered with polyphosphates rendered harmless $(NaPO_3)_6$.

$$(NaPO_3)_6 + Ca^{2+}/Mg^{2+} \longrightarrow \underset{\text{Soluble complex (Chelate)}}{Na_4CaP_6O_{18}(aq)} + 2Na^+$$

* Degree of Hardness :

It is defined as amount of $CaCO_3$ equivalent to amount all hardness causing substance present in 10^6 g of water. Its unit is ppm (parts per million)

eg. 100 ml water (hard)

gm

- x g $CaSO_4$ (aq), ✓ (Yes it causes hardness)
- y g KCl (aq), X
- z g $Mg(HCO_3)_2$ (aq) ✓
- p g $MgCl_2$ (aq) ✓

$$\left\{ \frac{x}{(40+96)} \times 2 + \frac{z}{(24+2\times 61)} \times 2 + \frac{p}{(24+71)} \times 2 \right\} \times \frac{10^6}{10^2}$$

= No. of g.eq of $CaCO_3$ *.

$$\Rightarrow \frac{W_{CaCO_3}}{100} \times 2 = \left\{ \frac{x}{136} \times 2 + \frac{z}{(24+2\times 61)} \times 2 + \frac{p}{95} \times 2 \right\} \times \frac{10^6}{100}$$

$\Rightarrow$ W_{CaCO_3} = degree of hardness.

Q. A certain sample of hard water is found to contain 68 ppm of $CaSO_4$ and 19 ppm of $MgCl_2$. Find degree of hardness of water.

Soln: In 10^6 g of hard water

- 68 g of $CaSO_4$, ✓
- 19 g of $MgCl_2$, ✓

$$\Rightarrow \left\{ \frac{68}{136} \times 2 + \frac{19}{95} \times 2 \right\} = \frac{W_{CaCO_3}}{100} \times 2$$

$$W_{CaCO_3} = 200 \times (0.5 + 0.2) \times 100$$

$$= 70 \text{ ppm.}$$

gmf

Q. What mass of quick lime is required to removed hardness of $1m^3$ of water which contains 1.62 mg of $Ca(HCO_3)_2$ per lit. of this hard water.

Soln: $Ca(HCO_3)_2(aq) + CaO(s) \longrightarrow 2CaCO_3\downarrow + H_2O(l)$

$1.62 \times 10^{-3} g \times 10^3$ ↑ 10^{-2} mol

$= \frac{1.62 g}{162}$ $= 0.01 \times 56$

$= 10^{-2}$ mol $= 0.56 g$ Ans.

12×10^{-5} % of $MgSO_4$ and 11.1×10^{-5} % $CaCl_2$ is present in water. Find measured hardness of water. Find no. of millimoles of washing soda (Na_2CO_3) to purify $1m^3$ of water.

Sol^n :

$$\left\{ \frac{12 \times 10^{-5}}{120} \times 2 + \frac{11.1 \times 10^{-5}}{111} \times 2 \right\} \times \frac{10^6}{100} = \frac{W_{CaCO_3}}{100} \times 2$$

$$W_{CaCO_3} = 2 \times 10^{-6} \times 10^6$$

$$= 2 \text{ ppm} = \text{degree of hardness.}$$

0.01 mol

$$MgSO_4 + Na_2CO_3 \longrightarrow MgCO_3 \downarrow + Na_2SO_4$$

$$CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3 \downarrow + 2NaCl$$

$$\frac{11.1 \times 10^{-5}}{100} \times \frac{10^6}{111} \text{ mol}$$

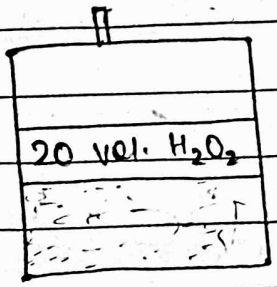
$$= 0.01 \text{ mol}$$

$$(n_T)_{Na_2CO_3} = 0.02 \text{ mol}$$

$$= 20 \text{ mmol} \quad \text{Ans}$$

* Volume Strength of H_2O_2 Solution :

A concentration term used for H_2O_2 solution.



↳ 'V vol H_2O_2 soln' means 1 lit of this H_2O_2 solution is capable of giving a maximum of V lit of O_2 gas under STP.

$$H_2O_2(aq) \longrightarrow H_2O(l) + \frac{1}{2}O_2(g)$$

1 lit

$n_{H_2O_2} = 2 \times \frac{V}{22.4}$ mol

$= \frac{V}{11.2}$ mol.

$\frac{V}{22.4}$ mol of O_2

Molarity of H_2O_2 soln $= \frac{(V/11.2)\ mol}{1\ lit}$

$$M_{H_2O_2} = \frac{V}{11.2}$$

$$N_{H_2O_2} = \frac{V}{11.2} \times 2$$

$$N_{H_2O_2} = \frac{V}{5.6}$$

v_f of H_2O_2 as OA & RA $= 2$

$H_2O_2 \xrightarrow{O.A} O_2$ (-1)

(R.A) ↑ $v_f = 2$

$H_2O_2 \xrightarrow{RA} H_2O$ (-1) (-2)

(O.A) ↑ $v_f = 2$

Q. What volume of 0.4M H_2O_2 can be oxidised by 50 ml of 0.1M KMnO4 in acidic medium.

Sol^n $H_2O_2(aq) + KMnO_4 \xrightarrow{H^+} Mn^{2+} + O_2$

$V_f = 2$; $V_f = 5$

$V \times 0.4 \times 2 = 50 \times 0.1 \times 5$

$V = \frac{250}{8}$ ml Ans.

Q. H_2O_2 , V = ? , 5.6 Vol H_2O_2

+ KMnO4 50ml, 0.1M

$V \times \left(\frac{5.6}{11.2}\right) \times 2 = 50 \times 0.1 \times 5$

V = 25ml

Oleum : A solution of SO_3 in H_2SO_4

$SO_3 + H_2SO_4$ → oleum

Oleum : $H_2S_2O_7$
(Pyrosulphuric Acid)
1:1 molar mixture of SO_3 and H_2SO_4

↳ X % oleum (or an oleum labelled as X %)
this means that when (X − 100) g of water is mixed in 100 g of this oleum sample, entire SO_3 present in oleum is converted into H_2SO_4 and amount of pure H_2SO_4 obtained will be X g.

eg. 109 % oleum
when 9g of water is mixed in 100 g of this oleum sample, all SO_3 will be converted into H_2SO_4 and amount of H_2SO_4 = 109 g

→ $SO_3 + H_2O(l) \longrightarrow H_2SO_4$

$\frac{9}{18} = 0.5$ mol

0.5 mol

0.5 × 80

= 40g SO_3

100g, 109 % oleum → 40g of SO_3
→ 60g of H_2SO_4

→ $SO_3 + H_2SO_4$

free SO_3 — combined SO_3

Composition of Oleum :

100g sample of X % oleum,

$SO_3 + H_2SO_4$ 100g

$SO_3 + H_2O(l) \longrightarrow H_2SO_4$

$\frac{X-100}{18}$ mol

$\frac{X-100}{18} \times 80$ g

$\Rightarrow W_{SO_3} = \frac{(X-100)}{9} \times 40$ g

% of SO_3 by mass

$= (X-100) \times \frac{40}{9}$ %

% of H_2SO_4 by mass

= 100 − % of SO_3

* Range of Oleum rating :

X % oleum

% SO_3 by mass $= \frac{(X-100) \times 40}{9}$

100g sample of % X % oleum

$SO_2 + H_2SO_4$

$(X-100) \times \frac{40}{9}$ g

for maximum X, $(X-100) \times \frac{40}{9} = 100$

$$X_{max} = 122.5$$

for minimum X, $(X-100) \times \frac{40}{9} = 0$

$$X_{min} = 100$$

gmp

$$100 < X < 122.5$$

gmp Q. In 100 g of 109% oleum is mixed with 4.5 g of water. Find new rating of oleum.

Soln: (104.5) g oleum Sample can still add 4.5 g of H_2O to convert SO_3 into H_2SO_4

104.5 g oleum ——— = 4.5 g

∴ 100 g oleum ——— = $\frac{4.5}{104.5} \times 100$

New oleum rating = $100 + \frac{4.5}{104.5} \times 100 = 104.31\%$

Q. 100 g 118 % oleum sample is taken.

Find (i) mass of SO_3 present in this sample
(ii) what volume (in lit) of 1M NaOH is needed to react or neutralize completely this oleum.

(i) $\% SO_3 = \frac{(118-100) \times 40}{9}$

$= 80\%$

$\boxed{W_{SO_3} = 80g}$

(ii) $\underset{(aq)}{H_2SO_4} + 2NaOH\,(aq) \longrightarrow Na_2SO_4\,(aq) + 2H_2O$

$\frac{118}{98}$ mol

$2 \times \frac{118}{98} = V \times 1 \Rightarrow V_{NaOH} = \frac{118}{49}$ lit